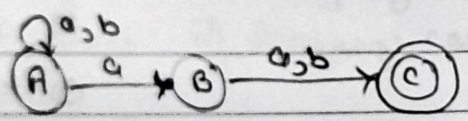
Q.

	a	b
→ A	{A, B}	{A}
* B	—	—

⇓ DFA

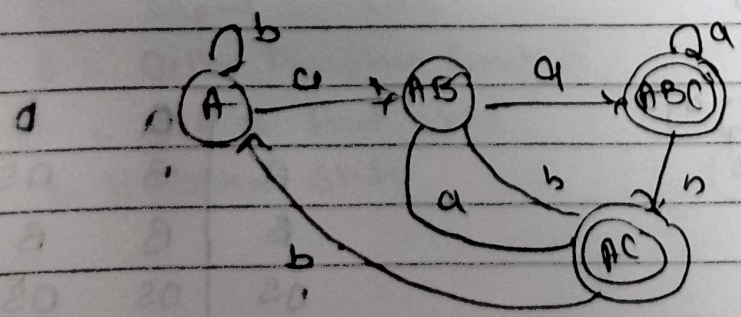
	a	b
A	AB	A
→ B	DS	DS
* DS	DS	DS
AB	AB	A

Q



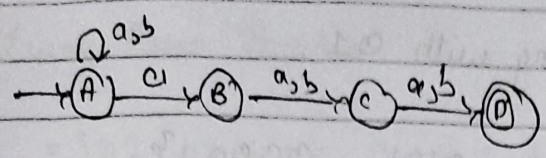
	a	b
A	{A, B}	{B}
B	{C}	{C}
C	—	—

⇓ DFA



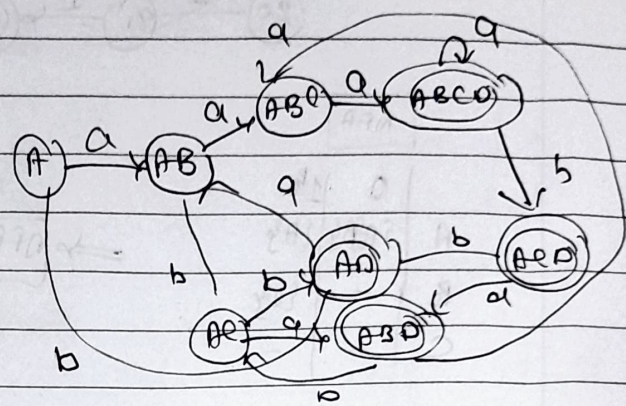
	a	b
A	AB	B
B	C	C
AB	ABC	AC
AC	AB	A
ABC	ABC	AC

Q.

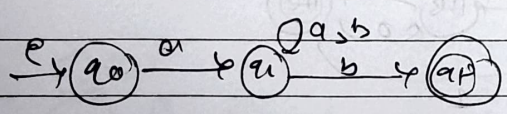


	a	b
A	AB	A
B	C	C
AB	ABC	AC
ABC	ABCD	ACD
AC	AB	A
ABCD	ABCD	ACD
ACD	ABD	AD
ABD	ABC	AC

	a	b
A	{A,B}	{A}
B	{C}	{C}
C	{D}	{D}
D	—	—

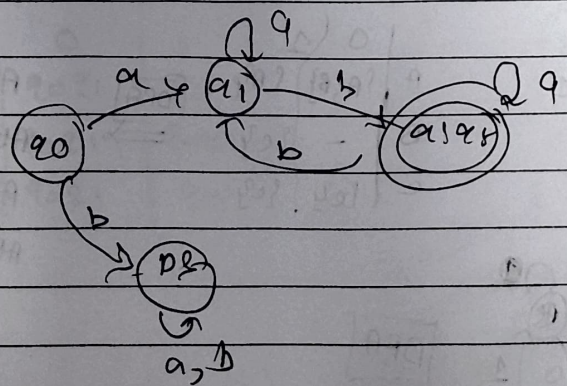


Q. Conversion starts with a & ends with b



	a	b
q0	{q1}	—
q1	{q1}	{q1, q4}
q4	—	—

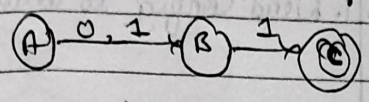
⇓



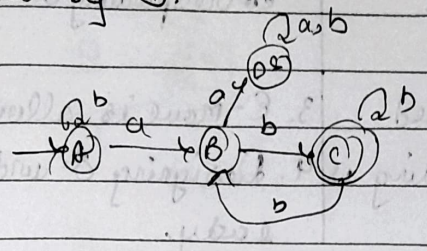
	a	b
q0	q1	D
q1	q1	q1q4
q1q4	q1	q1q4
D	—	—

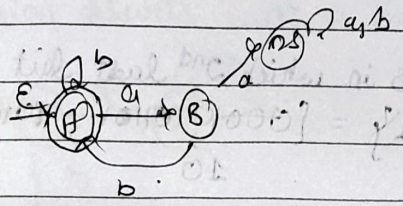
Q. Every Even Position 1

= {01, 1, 01, 1, 01, }



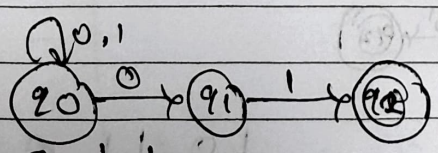
Q. A followed by B.





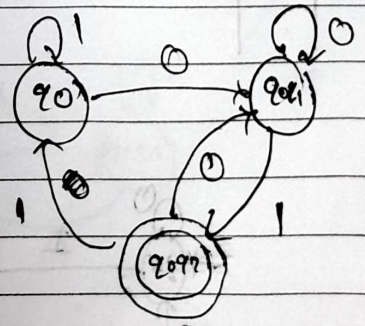
Practice Question

01 = L = {0101, 01, 0001, 1001, 1101, ...}

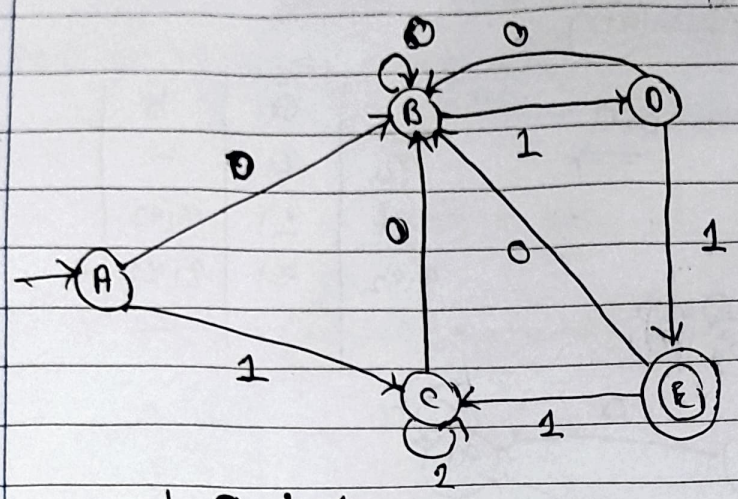


	0	1
q0	q0q1	q0
q1	—	q2
q2	—	—

	0	1
q0	q0q1	q0
q0q1	q0q1	q0q2
q0q2	q0q1	q0



Minimization of DFA :-



	0	1
→ A	B	C
B	B	1
C	B	C
D	B	E
(E)	B	C

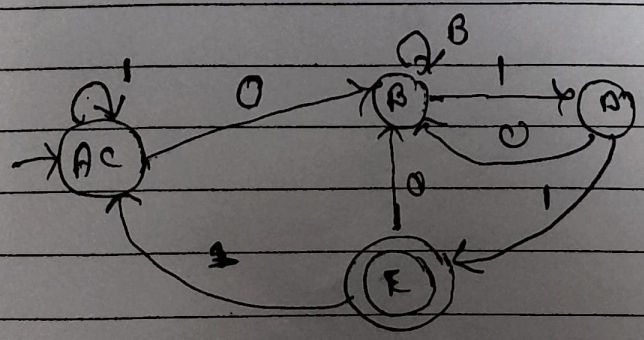
Equivalence ⇒
[non-final state together & fi state together.

0 Equivalence : {A, B, C, D} {E}

1 Equivalence : {A, B, C} {D} {E}

2 Equivalence :
{A, C} , {B} {D} {E}

3 Equivalence :
{A, C} {B} {D} {E}

For 0 & 1
[A, B] - [C, D]
They fall in same s
so they are equiv

[A, C] ✓
[C, D] ✗